

Three Geometries

Let V be a real vector space of dimension n .

The Geometry of Lengths and Angles (Inner Product/Riemannian Geometry)

Defn

An inner product (in the context of this talk, also called a metric): is a map $g : V \times V \rightarrow \mathbb{R}$ satisfying:

- (1) Bilinearity: $g(ax + y, z) = ag(x, z) + g(y, z)$.
- (2) Symmetry: $g(x, y) = g(y, x)$.
- (3) Positive-definiteness: $g(x, x) \geq 0$ for all $x \in V$, with equality if and only if $x = 0$.

Example

Dot product.

Why do we care about inner products? They give us the geometry of lengths and angles: Given a vector $v \in V$, its length is $|v| = \sqrt{g(v, v)}$, and the angle between two vectors is defined to be $\theta = \arccos \left(\frac{g(u, v)}{|u||v|} \right)$.

Example

With the dot product, our formula for length is the length of a vector by the Pythagorean theorem. Why is the formula for the angle valid? The law of cosines tells us $|u - v|^2 = |u|^2 + |v|^2 - 2|u||v| \cos \theta$, which by polarization $g(u, v) = \frac{1}{4}(|x + y|^2 - |u - v|^2)$ is equivalent to $g(u, v) = |u||v| \cos \theta$.

The dot product is the most important inner product, for the following reason:

Theorem (Corollary of Gram-Schmidt)

There exists a basis (f_i) for V such that

$$g(x^i f_i, y^i f_i) = \sum_i x^i y^i$$

So any inner product g looks like the dot product after choosing our basis.

So inner products give us one geometry, the geometry of lengths and angles.

The Geometry of Conservation

Inner products might be familiar territory, this one might be weirder.

Defn

The dual of V is the vector space V^* of linear maps $V \rightarrow \mathbb{R}$.

Defn

A symplectic form on V is a function $\omega : V \times V \rightarrow \mathbb{R}$ satisfying:

- (1) $\omega(u, v) = -\omega(v, u)$ (Alternating/Skew-Symmetric)
- (2) $\omega(ax + y, z) = a\omega(x, z) + \omega(y, z)$ (Bilinearity)
- (The above 2 conditions make ω a 2-form)
- (3) The linear map $V \rightarrow V^*$ defined by $v \mapsto \omega(v, \cdot)$ is injective. (Nondegenerate)

Example

Consider $V = \mathbb{R}^{2n}$, and let our coordinates be $(x_1, y_1, \dots, x_n, y_n)$. We're going to use some weird notation here:

Define $dx_i : V \rightarrow \mathbb{R}$ that takes the x_i th coordinate of a vector, so $dx_i \in V^*$. Then, define the product of dx_i and dy_i to be $dx_i dy_i$ defined by $dx_i dy_i(u, \cdot)$ to be the linear map $dx_i(u)dy_i(\cdot) - dx_i(\cdot)dy_i(u)$. Then, define

$$\omega = \sum_i dx_i dy_i$$

This is a symplectic form: any individual $dx_i dy_i$ is a 2-form, we need the sum for nondegeneracy.

Ok, why do we care at all about symplectic forms? Here's one version:

Example

In $\mathbb{R}^2$, $\omega(u, v) = \det[u|v]$. This is pretty nifty: we can think of ω above as determinants on the spans $\text{Span}\{x_i, y_i\}$. So 2-forms measure 2-dimensional oriented areas, the same way determinants do: we can think of them as the generalization of it.

But here's more of a reason:

Theorem (Noether's Theorem, loosely)

Suppose V is a space that obeys a "conservation law", that is, there is some reasonable function $H : V \rightarrow \mathbb{R}$ that measures a quantity, and that quantity is conserved, i.e., H doesn't change in the appropriate sense as the system evolves over time. Then, (V, H) is entirely determined by the calculus of some symplectic form ω . Conversely, a symplectic space (V, ω) gives a conservation law.

(specifically, a system of first order partial differential equations called Hamilton's equations) Why? Well, what does describing how the system changes over time look like? (Draw integrals and flow lines) These flow lines correspond to a vector field: an assignment of an arrow to each point. Thinking of the vector as an arrow, we obtain a function $X : V \rightarrow TV \cong V \times V$ (small handwave technicality) satisfying $X(v) = (v, \text{something})$.

Thus, we really want to find the component function $X_1 : V \rightarrow V$. Moreover, since we care about local evolving, we want something that only depends on the derivative dH of H . Fact: $dH = \frac{\partial H}{\partial x_i} dx_i$ is actually in V^* , so we want an isomorphism $V^* \rightarrow V$, or in other words an isomorphism $V \rightarrow V^*$, which satisfies $\omega(X, Y) = dH(Y)$.

This is precisely nondegeneracy, and we want bilinearity for technical reasons.

What about alternating? Well, we want H to not change as it evolves over time. So $dH(X) = 0$ (ALEX ACTUALLY EXPAND THIS TO SEE WHAT IT LOOKS LIKE ON THE INTEGRAL CURVES), so $\omega(X, X) = 0$, which is equivalent to alternating. There's also some calculus to be done, but we'll skip that. ■

In the same way as any inner product has a standard form, so too does any symplectic form:

Theorem (Linear Darboux's Theorem)

Let V be a vector space with a symplectic form ω . Then (V, ω) then there is a basis $(x_1, y_1, \dots, x_n, y_n)$ on V (so V is even dimensional) on which $\omega = \sum_i dx_i dy_i$. The proof is very similar to the proof of Gram-Schmidt.

We now have two geometries. What's the third?

Linear Complex Geometry

Every vector space over $\mathbb{C}$ is a real vector space, simply by forgetting that we can scalar multiply by complex numbers instead of just real ones. What about the opposite?

Definition

Let V be a real vector space. A (linear) complex structure on V , sometimes called an almost complex structure, is a linear map $J : V \rightarrow V$ satisfying $J^2 := J \circ J = -\text{Id}$. Such a J behaves like "multiplication

by i):

Proposition

Given the existence of such a J , V turns into a complex vector space denoted V_J with the same addition, and scalar multiplication:

$$(x + iy)v = xv + yJ(v)$$

Example

If we think about $\mathbb{R}^{2n}$ with basis $(x_1, y_1, \dots, x_n, y_n)$, the linear map J that satisfies $J(x_i) = y_i$, $J(y_i) = -x_i$ is a complex structure. If we let $z_i = x_i + iy_i$, then (z_i) is a complex basis of $\mathbb{R}^{2n}$ and J is literally multiplication by i . Thus, $\mathbb{C}^n \cong \mathbb{R}^{2n}$ as complex vector spaces, under these bases.

Theorem

Every vector space V with a complex structure J satisfies $V_J \cong \mathbb{C}^{\dim_{\mathbb{R}} V/2}$, where we can pick a basis $(x_1, y_1, \dots, x_n, y_n)$ of V such that J is like the above example.

Why do we care about almost complex structures? Other than the usual “we like complex numbers”, they linear-algebraify a lot of calculus:

Proposition

Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a differentiable function. Then f is holomorphic (under $\mathbb{C} \cong \mathbb{R}^2$) if and only if Df commutes with J .

Proof:

Since $J = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$, $DfJ = JDf$ becomes

$$\begin{bmatrix} (f_1)_y & -(f_1)_x \\ (f_2)_y & -(f_2)_x \end{bmatrix} = \begin{bmatrix} -(f_2)_x & -(f_2)_y \\ (f_1)_x & (f_1)_y \end{bmatrix}$$

the bottom row $(f_1)_x = (f_2)_y$, $(f_1)_y = -(f_2)_x$ is literally the Cauchy-Riemann equations. ■

2-out-of-3

We’ve seen we have a dot product, standard symplectic form, and standard complex structure on $\mathbb{R}^{2n}$. These are given by

$$J(u) = \begin{bmatrix} 0 & -\text{Id} \\ \text{Id} & 0 \end{bmatrix} u$$

$$\omega(u, v) = v^T \begin{bmatrix} 0 & -\text{Id} \\ \text{Id} & 0 \end{bmatrix} u$$

$$g(u, v) = v^T u$$

Hmm... that’s awfully nice. From this, you can check $\omega(u, v) = g(J(u), v)$.

Theorem (2-out-of-3)

Let V be a real vector space. Any of the following conditions imply the other two:

- (1) V has a symplectic form ω and almost complex structure J satisfying $\omega(J(u), J(v)) = \omega(u, v)$, $\omega(u, J(v)) > 0$ for all $u \neq 0$.
- (2) V has an inner product g and almost complex structure J satisfying $g(J(u), J(v)) = g(u, v)$.
- (3) V has a symplectic form ω and an inner product g .

Thus, we obtain a triple (g, ω, J) , called a compatible triple or a Kähler triple.

Proof:

If (1) holds, then $g(u, v) := \omega(u, J(v))$ works.

If (2) holds, then $\omega(u, v) := g(J(u), v)$ works.

(3) holding is the hard one. Note that both $u \mapsto \omega(u, \cdot)$ and $u \mapsto g(u, \cdot)$ are isomorphisms $V \rightarrow V^*$. Thus, there is a linear map A such that $\omega(u, v) = g(A(u), v)$ for some linear map $A : V \rightarrow V$. A is skew-symmetric because

$$g(A^T u, v) = g(u, Av) = g(Av, u) = \omega(v, u) = -\omega(u, v) = \omega(-u, v) = g(-Au, v)$$

Thus, $A^T = -A$, so $(AA^T)^T = AA^T$ and $g(AA^T u, u) = g(A^T u, A^T u) > 0$ for $u \neq 0$, which tells us AA^T is positive-definite.

Therefore, by the spectral theorem, A diagonalizes with positive eigenvalues: $AA^* = B \text{diag}(\lambda_1, \dots, \lambda_{2n}) B^{-1}$. We define $\sqrt{AA^*} = B \text{diag}(\sqrt{\lambda_1}, \dots, \sqrt{\lambda_{2n}}) B^{-1}$, which is still symmetric and positive-definite. Let $J = (\sqrt{AA^T})^{-1} A$ (the inverse exists since all eigenvalues are positive).

Then, J is orthogonal: $JJ^T = \text{Id}$, and it is skew-symmetric, $J^T = -J$ (annoying but easy to check from this formula), so $J^2 = -\text{Id}$, so J is a complex structure.

Then, it satisfies

$$\omega(J(u), J(v)) = g(AJu, Jv) = g(JAu, Jv) = g(Au, v) = \omega(u, v)$$

, since J commutes with A (by definition of the square root, since AA^T does), and $g'(u, v) := \omega(u, Jv)$ is a compatible inner product. ■

Corollary (2-out-of-3, Lie group version)

Let $O(2n)$ be the group of matrices A acting on $\mathbb{R}^{2n}$ such that $g(Au, Av) = g(u, v)$ for g the dot product, let $Sp(2n)$ be the group of matrices A acting on $\mathbb{R}^{2n}$ such that $\omega(Au, Av) = \omega(u, v)$ for ω the standard symplectic form, and $\text{GL}(n, \mathbb{C})$ the group of $n \times n$ complex invertible matrices, thought of as $2n \times 2n$ real matrices via the isomorphism $\mathbb{C}^n \cong \mathbb{R}^{2n}$, and let $U(n)$ be the group of $n \times n$ complex matrices that satisfy $AA^* = A^*A = \text{Id}$. Then,

$$U(n) = O(2n) \cap Sp(2n) = O(2n) \cap \text{GL}(n, \mathbb{C}) = \text{GL}(n, \mathbb{C}) \cap Sp(2n)$$